

Performance & Cost Optimization

Medical Document Parser - RAG Pipeline

Current: **\$8,800 / 100 providers | ~2 hours processing**

Target: **\$1,200 - \$2,100 / 100 providers | ~30-40 min processing**

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1. Current Cost Breakdown (Per 100 Providers)

97% of cost is GPT extraction tokens. All optimization must target GPT call reduction.

Component	Usage / Provider	Cost / Provider	Cost / 100	% Total
GPT Extraction (34 fields x 20 docs)	~3,400,000 tokens	\$85.00	\$8,500	97%
GPT Evaluation (6 metrics)	~68,000 tokens	\$2.00	\$200	2%
OCR (Document Intelligence)	~87 pages	\$0.87	\$87	1%
Embeddings	~68,000 tokens	\$0.01	\$1	<1%
AI Search	~34 queries	\$0.001	\$0.10	<1%
TOTAL		\$88.00	\$8,800	100%

Why cost is high: Each of 34 fields makes a separate GPT call per document. 20 documents x 34 fields = 680 GPT calls per provider. ~70% of these calls return empty because the document doesn't contain that field type.

2. Option A: Configuration Changes Only (No Code)

Constraints: Multimodal mode required (scanned docs). Single field per call required (accuracy). No batch API (crashes on 3000 docs).

#	Change	Config Setting	Cost Impact	Speed	Accuracy	Works?
1	Workers: 20	parallel_workers: 20	Same	25% faster	Same	YES
2	TPM: 5M	Azure Portal	Same (no retries)	No 429 errors	Same	YES
3	Content filter: Allow	Azure Portal	Same (no blocks)	No rejections	Same	YES
4	Top K: 2 chunks	rag.top_k: 2	-30% prompt tokens	Slightly faster	~2% drop	TEST
5	Eval off for bulk	evaluation.enabled: false	-100% eval cost	30% faster	No scores	YES
6	Text mode	rag.mode: text	-60% tokens	6x faster	DROPS	NO
7	Fields per call	fields_per_call: 3	-40% tokens	65% faster	DROPS	NO
8	Batch API	apitype: batch	-50%	Async	CRASHES	NO

Config-Only Combined Impact:

Scenario	Cost / 100	Savings	Time	Effort
Current baseline	\$8,800	-	~2 hours	-
Level 1: workers:20 + TPM:5M + filter	\$8,800	0%	~1.5 hours	5 minutes
Level 2: + eval off	\$8,600	2%	~1.2 hours	1 minute
Level 3: + top_k:2 (if accuracy holds)	\$6,000	32%	~1 hour	1 minute

Honest answer: Config-only saves 0-32%. Real savings require code changes.

3. Option B: Architecture & Code Changes

All techniques below maintain accuracy. No accuracy compromise.

#	Technique	How It Works	Cost	Accuracy	Effort
1	Document Classification	Classify doc type first. Extract only relevant fields (8 instead of 34)	-76%	Same/better	1 week
2	Response Caching	Same RAG chunk + same field = reuse previous GPT result	-20-30%	Same	2 days
3	Skip Duplicate Docs	Hash documents. Skip if same file already processed	-10-30%	Same	1 day
4	Prompt Compression	Shorter system prompt. Same meaning, fewer tokens	-20%	Same	2 days
5	Embedding Cache	Cache 34 field embeddings. Reuse across all documents	-5%	Same	1 day
6	Smaller Model Routing	GPT-4o-mini for name/email. GPT-5 for licenses	-40%	~1% drop simple	2 days
7	Document Filtering	Skip blank/tiny/irrelevant docs before OCR	-10-20%	Same	1 day
8	Provider-Level Parallel	Process 3 providers simultaneously	Same	Same	2 days

Code Change Combined Impact:

Phase	What to Build	Cost / 100	Savings	Time	Dev Effort
Current	Nothing	\$8,800	-	~2 hours	-
Phase 1	Config: workers:20, TPM:5M, eval off	\$8,600	2%	~1.5 hours	5 min
Phase 2	+ Document Classification	\$2,100	76%	~40 min	1 week
Phase 3	+ Response cache + skip dups	\$1,500	83%	~35 min	+ 3 days
Phase 4	+ Prompt compression + embed cache	\$1,200	86%	~30 min	+ 3 days
Phase 5	+ Smaller model routing	\$750	91%	~25 min	+ 2 days

4. Document Classification - Detailed Explanation

The single biggest cost reduction (76%) with zero accuracy loss. This is the #1 priority.

4.1 The Problem: Wasted GPT Calls

Currently, every document is processed against all 34 fields. But a license document only contains license information. Running 34 fields on it means 26 calls return empty. That's 26 wasted GPT calls per document.

Document	Fields Extracted	Fields Empty (Wasted)	Waste %
License.pdf	8 license fields found	26 fields return empty	76%
DEA Certificate.pdf	4 DEA fields found	30 fields return empty	88%
BLS Card.pdf	2 BLS fields found	32 fields return empty	94%
CV/Resume.pdf	10 personal fields found	24 fields return empty	71%
Vaccine Record.pdf	6 vaccine fields found	28 fields return empty	82%
Insurance Form.pdf	8 insurance fields found	26 fields return empty	76%
AVERAGE	~8 fields per doc	~26 wasted calls per doc	~76%

Per provider (20 docs): 20 x 26 wasted calls = 520 wasted GPT calls = \$65 wasted per provider

4.2 The Solution: Classify First, Then Extract

Add one cheap classification call per document. Then only extract relevant fields for that document type.

Step 1: After OCR, send first 500 characters to a cheap model (GPT-4o-mini):

Prompt: "Classify this document into one category: license, dea, board_cert, bls_cert, vaccine, insurance, personal, other. Return category name only."

Cost: ~100 tokens x \$0.00015 = \$0.000015 per document (basically free)

Step 2: Map document category to relevant fields:

Document Category	Fields to Extract	Count	Example Documents
license	license_number, license_state, license_issue_date, license_expiration_date, license_state_license_and_medical_license	5	State License, Medical License, FSMB

4.3 How Classification Works in the Pipeline

The classification step adds ~0.5 seconds per document (one cheap GPT call). It saves ~26 GPT-5 calls per document (~\$3.25 saved per document).

Current Pipeline:

Document → OCR → Chunk → Index → Extract ALL 34 fields → Aggregate → Output

New Pipeline (with classification):

Document → OCR → **CLASSIFY (1 cheap call)** → Chunk → Index → Extract ONLY relevant fields → Aggregate → Output

4.4 Implementation Details

New file: services/document_classifier.py

Config addition:

```
"classification": {"enabled": true, "model": "gpt-4o-mini", "categories": {...}}
```

Changes to: utils/provider_processor.py (add classification call before extraction)

No changes to: extraction engine, aggregation, evaluation, output, prompts

Accuracy impact: NONE. Classification doesn't change how fields are extracted. It only skips fields that would return empty anyway. A license document never has vaccine data. Skipping vaccine fields on a license document doesn't lose any information.

Fallback safety: If classification returns 'other' or fails, ALL 34 fields are extracted (current behavior). No data is ever lost.

5. Side-by-Side Comparison

Metric	Current	Config Only (Option A)	Code Changes (Option B)	Combined (A + B)
Cost / 100 providers	\$8,800	\$6,000 - \$8,600	\$750 - \$2,100	\$750
Cost savings	-	2 - 32%	76 - 91%	91%
Processing time	~2 hours	~1 - 1.5 hours	~25 - 40 min	~25 min
Speed improvement	-	1.3 - 2x	3 - 5x	5x
GPT calls / document	34	34	4 - 8	4 - 8
Accuracy impact	Best	0 - 2% drop	Zero loss	Zero loss
Development effort	-	0 days	2 - 3 weeks	3 weeks
Risk level	-	Very low	Low	Low

6. Annual Cost Projection (4 Runs / Month)

Scenario	Cost / Run	Monthly	Annual	Annual Savings
Current baseline	\$8,800	\$35,200	\$422,400	-
Config only (Level 3)	\$6,000	\$24,000	\$288,000	\$134,400
+ Document Classification	\$2,100	\$8,400	\$100,800	\$321,600
+ Response cache + skip dups	\$1,500	\$6,000	\$72,000	\$350,400
+ All code optimizations	\$750	\$3,000	\$36,000	\$386,400

Document classification alone saves \$321,600 per year.

7. Recommendation

Immediate (Today): Apply config changes - workers:20, TPM:5M, content filter:Allow. Cost: same. Speed: 25% faster.

Week 1-2: Implement Document Classification. Cost drops 76% (\$8,800 → \$2,100). Zero accuracy loss. 1 week development.

Week 3-4: Add response caching + duplicate document skipping. Cost drops to \$1,500. 3 days development.

Month 2: Prompt compression + embedding cache + smaller model routing. Cost drops to \$750.

Bottom Line: Document classification is the single most impactful optimization. It reduces cost by 76% with zero accuracy loss by eliminating 520 wasted GPT calls per provider. Annual savings: \$321,600. Development effort: 1 week.